

Universidad Ana G. Mendez Recinto Gurabo

COMP315

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Final Proyect

Movie Database

Introduction

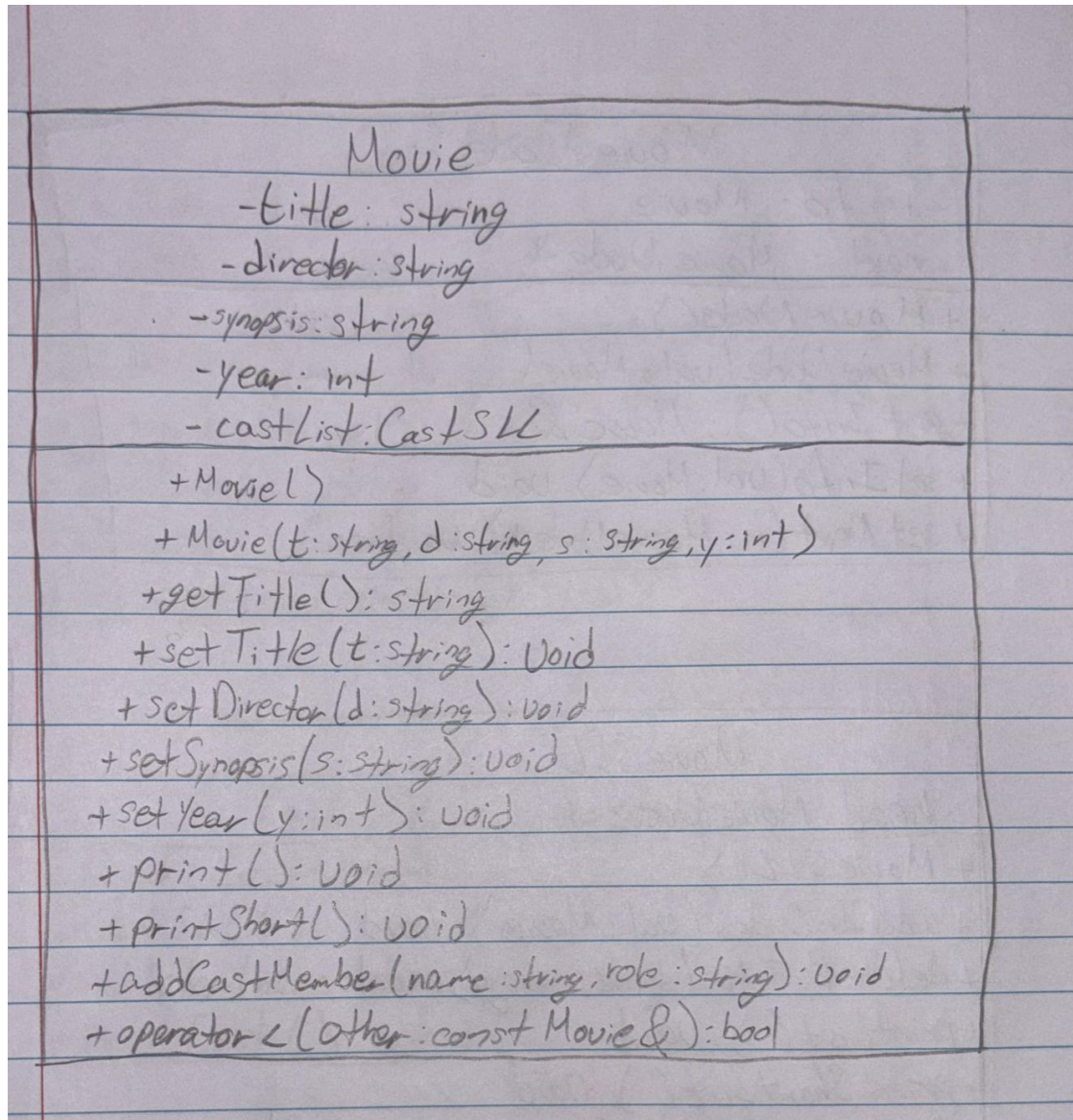
The Movie Database Management System is a specialized C++ application designed to organize and manage large collections of films efficiently. The program utilizes a **Doubly Linked List (DLL)** to manage movie categories or genres, allowing for easy bidirectional navigation. Within each genre, the system employs **nested Singly Linked Lists (SLL)** to manage specific movie entries and their respective cast members. This hierarchical approach ensures that data is stored logically while maintaining the flexibility of dynamic memory allocation.

System Architecture

The following UML diagrams represent the blueprint of the system. They illustrate how the GenreDLL acts as the primary container, holding GenreNode objects that each link to a MovieSLL. This chain continues down to individual Movie objects and their associated CastSLL.

For the Cast & Crew data we used a Single Linked List to prioritize memory efficiency and linear simplicity. Also because on the class we did not discussed the BST (Binary Search Tree) before the date of April 28 2026.

UMLS



Movie Node

```
- info: Movie  
- next: Movie Node *  
+ MovieNode()  
+ MovieNode(val: Movie)  
+ get Info(): Movie &  
+ set Info(val: Movie): void  
+ set Next(n: Movie Node *): void
```

Movie SLL

```
- head: Movie Node *  
+ MovieSLL()  
+ add In Order (val: Movie): void  
+ delete By Title (title: string): bool  
+ print List(): void  
+ print Short List(): void  
+ find Movie By Title (title: const string &): Movie Node *  
+ get Head(): Movie Node *
```


Cast Node

```
- actorName: string  
- role: string  
- next: CastNode*  
+ CastNode(name: string, r: string)  
+ getActorName(): string  
+ getRole(): string  
+ getNext(): CastNode*  
+ setNext(n: CastNode*): void
```

Cast SLL

```
- head: CastNode*  
+ CastSLL()  
+ addActor(name: string, role: string): void  
+ printCast(): void
```

Genre Node

```
+ name: string  
+ movies: Movie SLL  
+ next: Genre Node *  
+ prev: Genre Node *  
+ Genre Node(n: string)
```

Genre DLL

```
- head: Genre Node *  
- tail: Genre Node *  
+ Genre DLL()   
+ addGenre(name: string): void  
+ printGenre(): void  
+ listMoviesByGenre(name: const string&): void  
+ searchMoviesByGenre(title: const string&): void  
+ modifyMovie(title: string): void  
+ findGenre(name: string): Genre Node *
```

Key Implementation Details

- **Alphabetical Insertion:** The `MovieSLL::addInOrder` method ensures that the database remains organized by comparing the title of a new movie with existing nodes using the overloaded `<` operator. It traverses the list until it finds the correct alphabetical position, ensuring $O(n)$ insertion that keeps the list sorted for display.
- **Tiered Data Structures:** To find a specific film, the logic first searches the `GenreDLL` to locate the correct `GenreNode`. Once the genre is identified, the program accesses the internal `MovieSLL` to perform operations like searching, deleting, or modifying movie details.
- **Dynamic Memory:** The system relies entirely on pointers and the new keyword to create nodes (`GenreNode`, `MovieNode`, `Cast`) at runtime. This allows the database to grow or shrink based on the user's needs without the limitations of fixed-size arrays.

Scenario A:

Adding genres

```
Welcome to the MOVIE DATABASE MANAGEMENT SYSTEM!

=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
1
Please enter a genre:
Action

Genre added successfully!

Press enter to return to the menu...█
```

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
1
Please enter a genre:
Sci-fi

Genre added successfully!

Press enter to return to the menu...█
```


=====

1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit

Enter your choice:

1

Please enter a genre:

Thriller

Genre added successfully!

Press enter to return to the menu...

Scenario B:

Adding movies

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
2
Available genres:
- Action
- Sci-fi
Select genre:
Sci-fi
      Movie Details
Enter title: The Matrix
Enter director: Lana Wachowski
Enter synopsis: A computer hacker discovers that reality is a simulated dream world controlled by machines and joins a rebellion to liberate humanity.
Enter year: 1999
Movie added successfully!

Enter Actors name:
Keanu Reeves

Enter Role:
Neo
Add another actor? (Y/N)
y

Enter Actors name:
Laurence Fishburne

Enter Role:
Morpheus
Add another actor? (Y/N)
y

Enter Actors name:
Carrie-Anne Moss

Enter Role:
Trinity
Add another actor? (Y/N)
n

Movie and Cast added successfully!

Press enter to return to the menu...
```

Press enter to return to the menu...

=====

1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit

Enter your choice:

2

Available genres:

- Action
- Sci-fi

Select genre:

Sci-fi

Movie Details

Enter title: Inception

Enter director: Christopher Nolan

Enter synopsis: A professional thief who steals secrets by entering people's dreams is offered a chance at redemption by performing "inception"—planting an idea rather than stealing one.

Enter year: 2010

Movie added successfully!

Enter Actors name:

Leonardo DiCaprio

Enter Role:

Dom Cobb

Add another actor? (Y/N)

y

Enter Actors name:

Joseph Gordon-Levitt

Enter Role:

Arthur

Add another actor? (Y/N)

y

Enter Actors name:

Elliot Page

Enter Role:

Ariadne

Add another actor? (Y/N)

n

Movie and Cast added successfully!

Press enter to return to the menu...

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
2
Available genres:
- Action
- Sci-fi
Select genre:
Action
    Movie Details
Enter title: The Dark Knight
Enter director: Christopher Nolan
Enter synopsis: Batman faces a new, chaotic threat in the Joker, who seeks to dismantle Gotham City's sense of justice and order.
Enter year: 2008
Movie added successfully!

Enter Actors name:
Christian Bale

Enter Role:
Bruce Wayne
Add another actor? (Y/N)
y

Enter Actors name:
Heath Ledger

Enter Role:
Joker
Add another actor? (Y/N)
y

Enter Actors name:
Gary Oldman

Enter Role:
James Gordon
Add another actor? (Y/N)
n

Movie and Cast added successfully!

Press enter to return to the menu... 
```

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
2
Available genres:
- Action
- Sci-fi
- Thriller
Select genre:
Thriller
    Movie Details
Enter title: Parasite
Enter director: Bong Joon-ho
Enter synopsis: An impoverished family infiltrates the lives of a wealthy household by posing as unrelated, highly qualified workers, leading to a dark collision of social classes.
Enter year: 2019
Movie added successfully!

Enter Actors name:
Song Kang-ho

Enter Role:
Kim Ki-taek
Add another actor? (Y/N)
y

Enter Actors name:
Choi Woo-shik

Enter Role:
Kim Ki-woo
Add another actor? (Y/N)
y

Enter Actors name:
Park So-dam

Enter Role:
Kim Ki-jung
Add another actor? (Y/N)
n

Movie and Cast added successfully!

Press enter to return to the menu...
```


Scenario C:

Showing all genres

Press enter to return to the menu...

=====

1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit

Enter your choice:

4

Showing all genres

- Action
- Sci-fi
- Thriller

Press enter to return to the menu...

Scenario D:

Deleting movie

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
5
Showing available genres:
- Action
- Sci-fi
- Thriller
Please select the genre of the movie you want to delete:
Thriller
Please select the movie to delete:
Parasite
Movie deleted successfully deleted

Press enter to return to the menu...
```

Showing the movie has been deleted

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
5
Showing available genres:
- Action
- Sci-fi
- Thriller
Please select the genre of the movie you want to delete:
Thriller
Please select the movie to delete:
Parasite
Movie deleted successfully deleted

Press enter to return to the menu...
```

Scenario E:

Showing all movies from the selected genre

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
6
Showing available genres:
- Action
- Sci-fi
- Thriller
Enter the genre: Action
Showing all the movies from the selected genre:
Movies in Action
Furiosa: A Mad Max Saga(2024) - Dir: George Miller
The Dark Knight(2008) - Dir: Christopher Nolan

Press enter to return to the menu...
```

Scenario F:

Searching for a movie (that exists)

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
7
Please select the movie to search
Inception

Movie found in genre Sci-fi
Title: Inception
Director: Christopher Nolan
Year: 2010
Synopsis: A thief who extracts secrets from dreams is offered a chance at redemption if he can successfully plant an idea in a target's mind.
Cast & Crew:
- Leonardo DiCaprio as Dom Cobb
- Joseph Gordon-Levitt as Arthur
- Elliot Page as Ariadne

Press enter to return to the menu...
```

Searching for a movie (that it does not exist)

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
7
Please select the movie to search
Cars
Movie not found.

Press enter to return to the menu...
```

Scenario G:

Selecting another number that is not 1-8 from the menu

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
9
Invalid choice!!!! Choose bewteen 1-8

Press enter to return to the menu...
```

Scenario H:

Exit of the Menu

```
=====
1. Add a genre
2. Add a movie
3. Modify a movie
4. List all genres
5. Delete a movie
6. List all movies from a selected genre
7. Search for a movie
8. Exit
Enter your choice:
8
Goodbye!
```

Conclusion:

Developing this system highlighted the importance of careful pointer management, especially when dealing with nested structures. One major challenge was ensuring that deleting a movie also properly managed the memory for its associated cast list to avoid memory leaks. By implementing modular classes and clear setter/getter methods, I was able to successfully link three different levels of data (Genres → Movies → Cast) into a cohesive, functional application.